

Patient Healthcare Details Revenue Cycle Analysis

1. Project Overview and Objective

Patient Healthcare Details Revenue Cycle Analysis project involves cleaning, transforming, and analysing raw data using **Excel** and creating an interactive **Power BI dashboard** to derive meaningful business insights.

The main objective is to demonstrate data pre-processing techniques using Excel and an interactive Power BI dashboard visualization to make informed decisions.

2. Data Sources

- **Source Description and Timeline:** <https://excelx.com/practice-data/generators/healthcare/> Dataset Search **from** JAN2024-JAN2025.
- **Domain:** Patient Healthcare Analysis.

3. Problem Statement

- Revenue leakage & claim performance
- Identify gaps between Amount Billed and Amount Paid to detect revenue leakage
- 2.track how often patients visit and through what channel.
- Which encounter types are most common and Which specialties are overloaded.
- 3.provider performance and workload distribution across specialties.
- 4.Comparison b/w operational and financial performance across healthcare facilities

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Patient _ ID	Integer / String	Unique identifier for each patient
Name /Provider and Insurance Provider	String (Text)	Name of the patient /Provider And Insurance Provider
Age	Numeric (Integer)	Age of the patient in years
Encounter Type/Specialty	Categorical	What kind of visit and specialty

5. Tools & Technologies

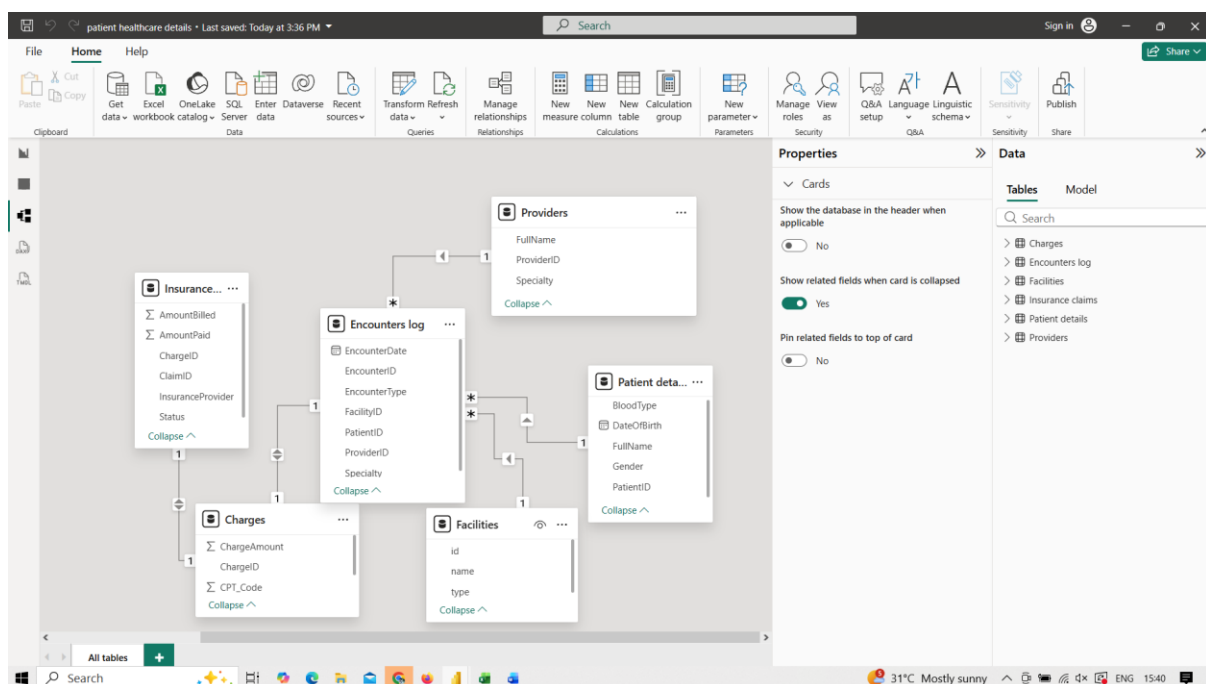
- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.

7.Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.

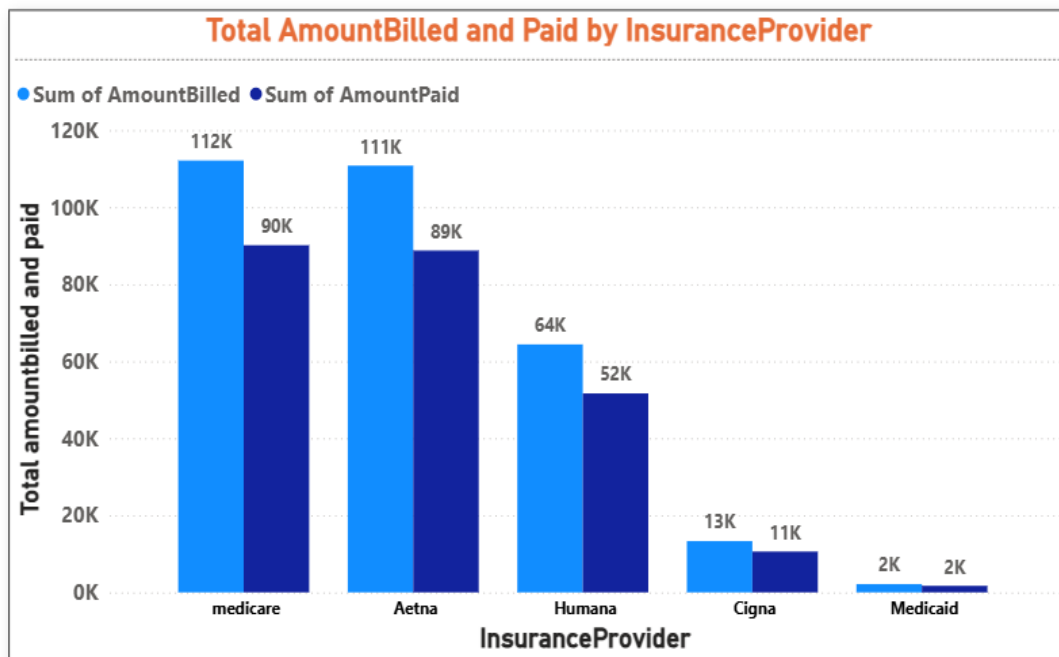


- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals.
 - Total _ paid = SUM('Insurance claims'[AmountPaid])
 - Min _ paidamount = MIN('Insurance claims'[AmountPaid])
 - Max _ paidamount = MAX('Insurance claims'[AmountPaid])
 - Total_ins_provider = COUNTROWS('Insurance claims')
 - Avg_chargeamt = AVERAGE(Charges[ChargeAmount])

8. Analysis and Visualizations (Power BI) Dashboard

Clustered column chart

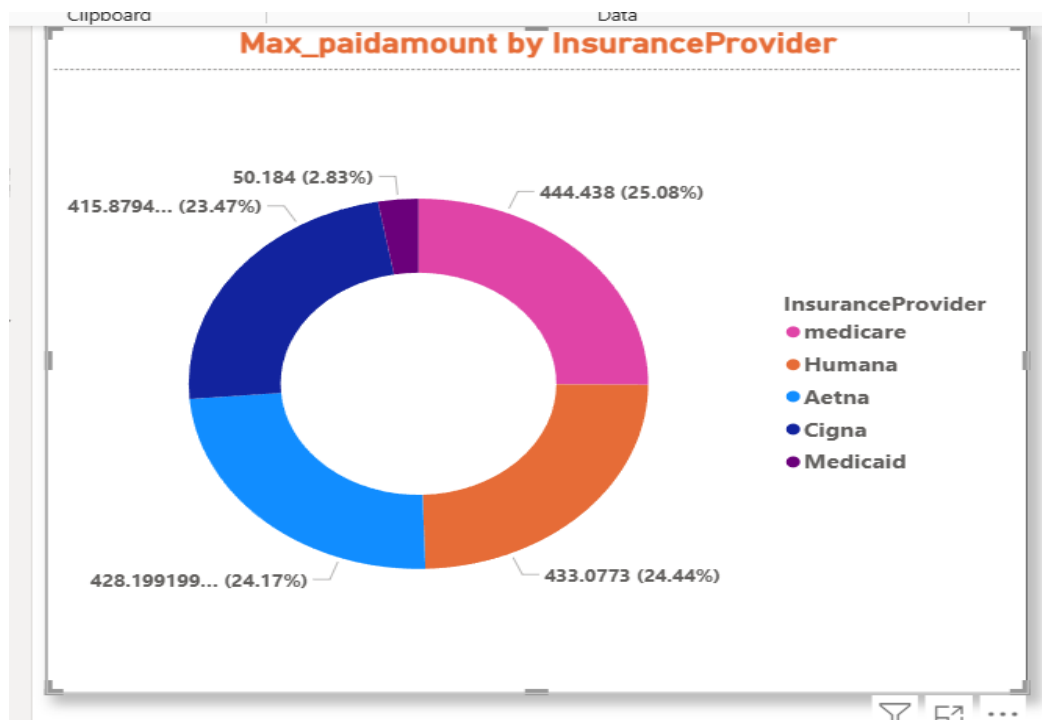
The chart compares **Amount Billed vs Amount Paid** by insurance provider.



Insights: The big players (Medicare & Aetna) drive the majority of revenue. comparatively revenue of Medicaid is very low. Medicare total billed amount 112k and paid amount 90k. Medicaid total billed amount 2145K and paid amount 1698k.

Donut chart:

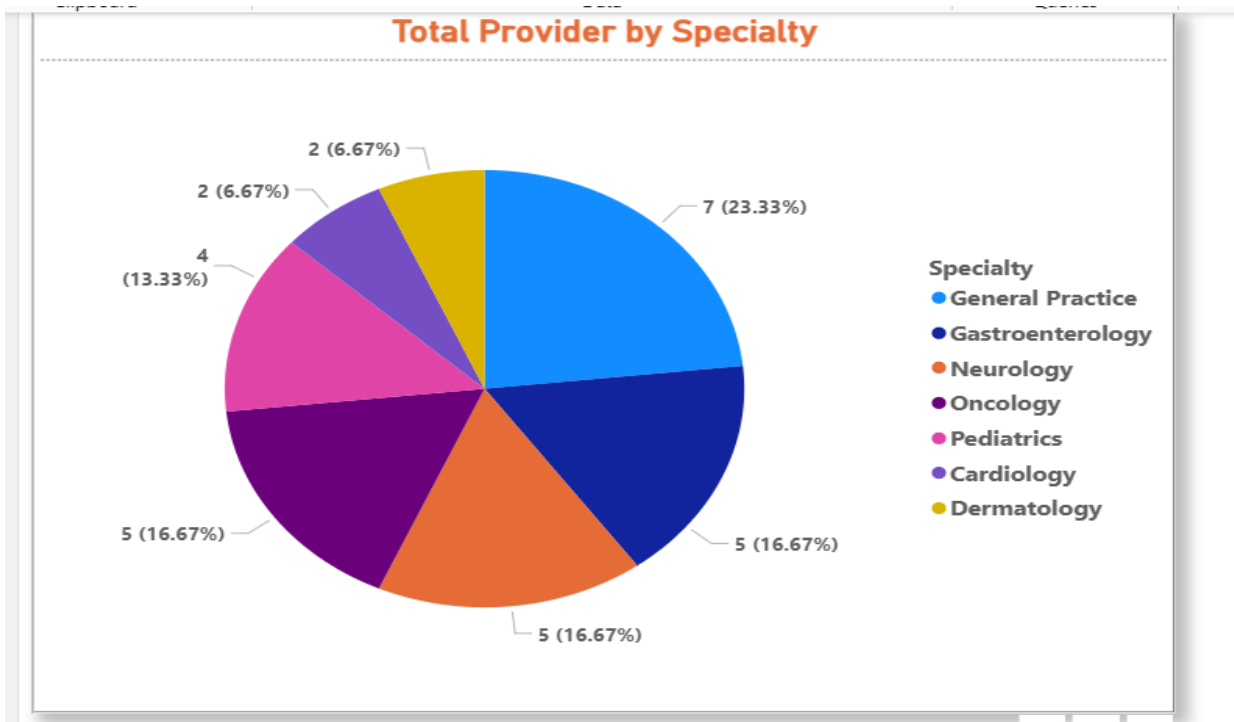
This chart provides billed revenue and most of the payment gap.



Insights: The major insurance providers Medicare account for most of the billed revenue. Humana and Aetna follow the same pattern at a smaller scale. Cigna and Medicaid contribute very little overall.

PIE CHART:

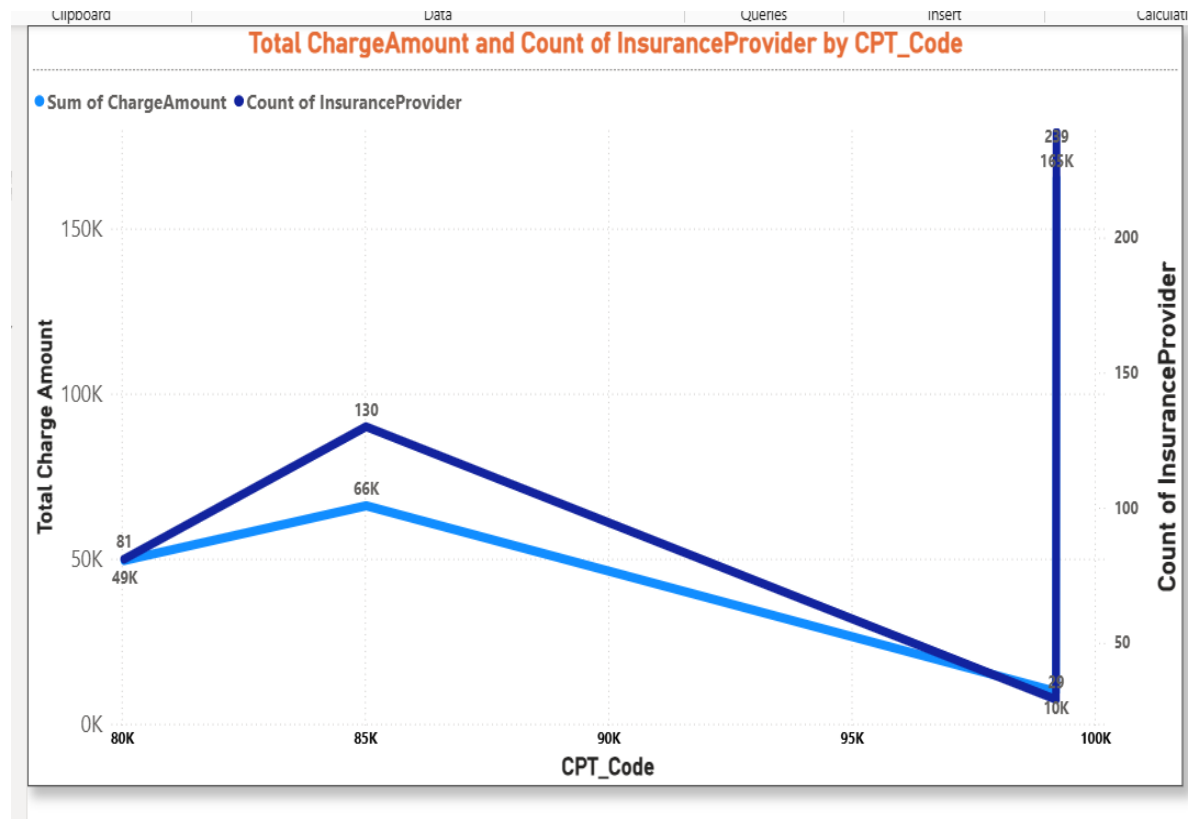
This chart shows **Total Providers by Specialty**.



Insights: General Practice slightly dominates. That makes sense operationally. Primary care usually feeds referrals into specialty care. It is your intake funnel. Cardiology and Dermatology are underrepresented. Only 2 providers each. That's capacity constraint territory.

Line chart:

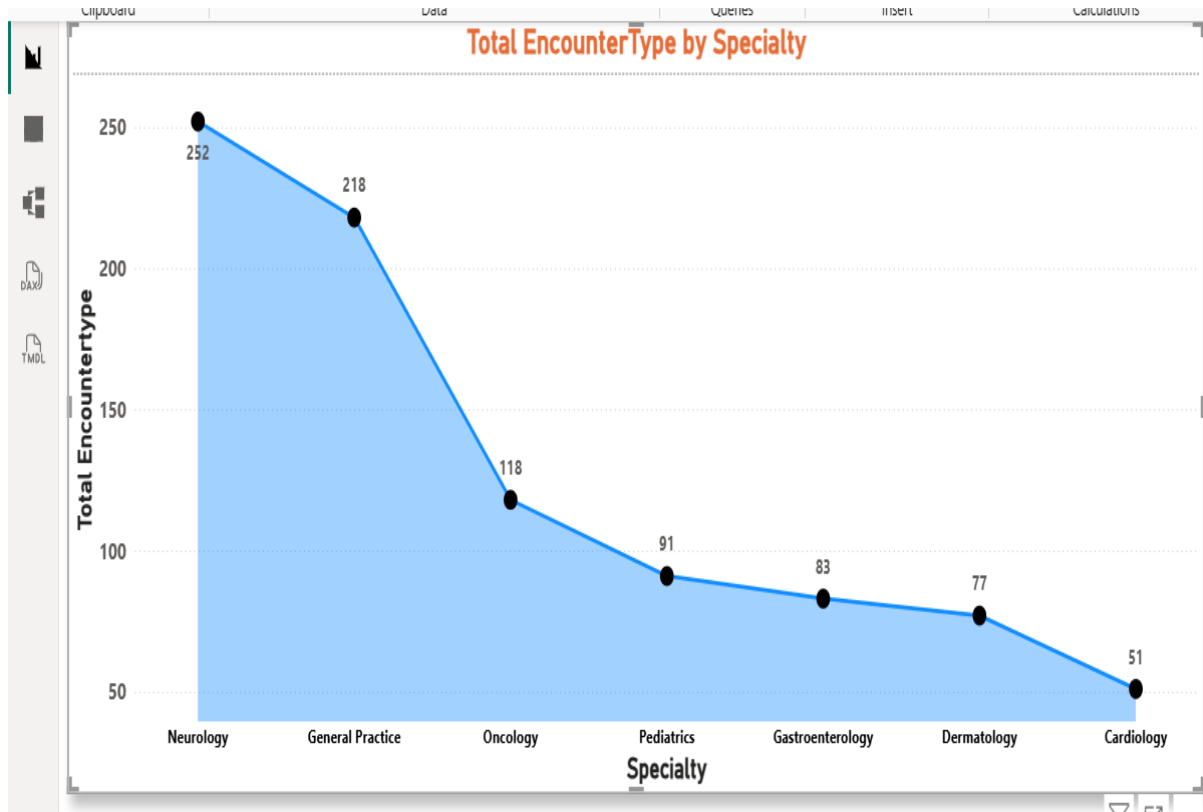
This chart shows **Total Charge Amount** and **Count of Insurance Provider** by **CPT_Code**.



Insights: In this chart use to find which CPT code having more charge amount and count of insurance providers. CPT 99213 total charge amount 166K this is the highest charge amount for all over the CPTs and 239 insurance providers . lowest charge amount 49k that CPT is 80053 and total insurance providers 81.

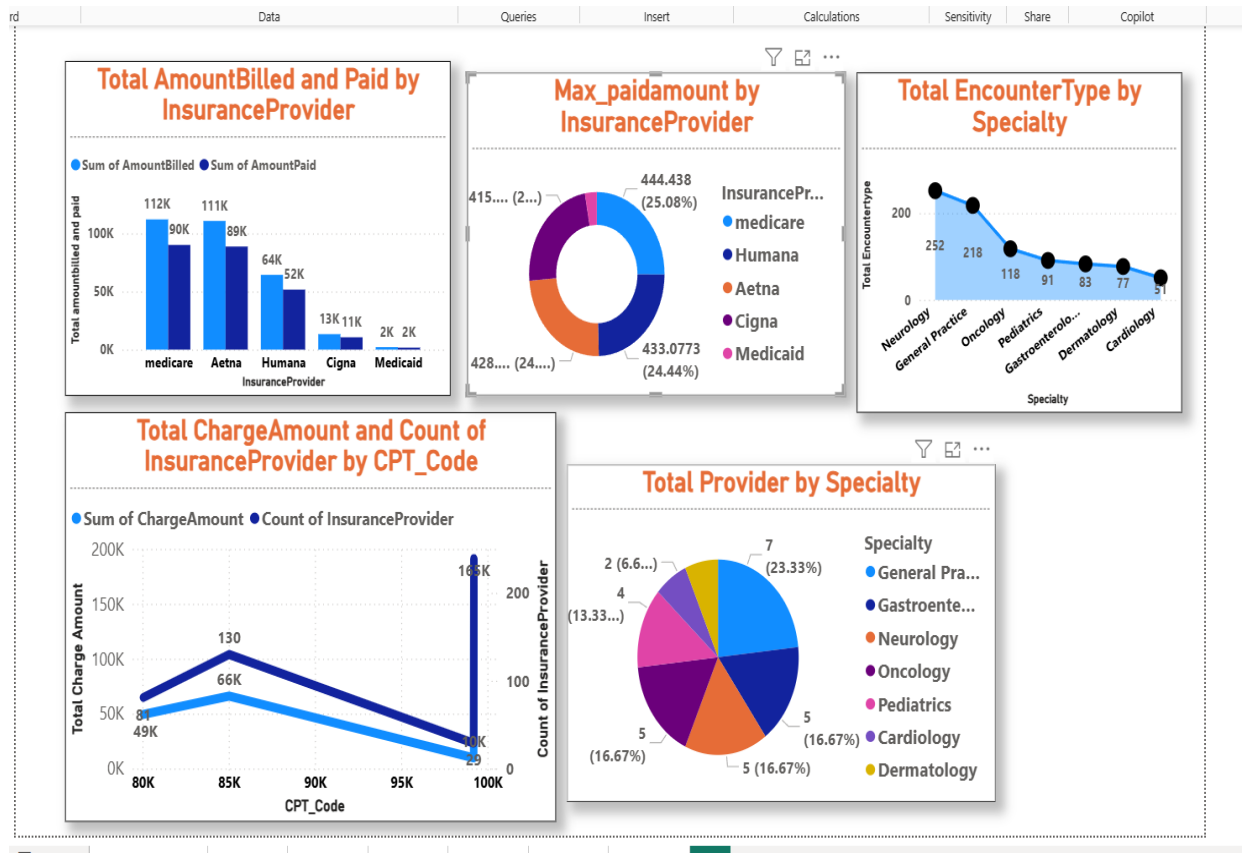
Area Chart:

This chart shows **Total Encounter Type by Specialty**.



Insights: Neurology is carrying the highest patient load. Totally 252 Encounters. Neurology has only 5 providers. General Practice is also heavy (218 encounters) and has 7 providers. Cardiology had only 2 providers and encounters 51. Neurology is overloaded and Cardiology is underloaded.

DASHBOARD /REPORT



9. Insights & Conclusions

- **Key Findings:**

- Revenue & Insurance Performance
- CPT Code Revenue Distribution
- Provider Distribution by Specialty
- Encounter Volume by Specialty

- **Provide the analysis insights:**

- Medicare and Aetna bring most of the revenue.
 - There is a gap between billed and paid amounts.
 - Neurology has the highest patient encounters.
 - General Practice has the most providers.
 - A few CPT codes generate most of the charges.

 - Payment gap exists because of insurance contracts and adjustments.
 - Revenue is concentrated in certain insurers and procedures.
 - Neurology has high encounters due to strong demand or referrals.
 - Some specialties have low encounters because demand is lower or referrals are weak.

 - Revenue risk is high if major insurers change policies.
 - Neurology providers may become overloaded.
 - If top CPT reimbursements drop, total revenue will fall.
 - Underutilized specialties may continue to underperform.

 - Monitor collection rate per insurer.
 - Improve negotiation with major payers.
 - Balance provider workload across specialties.
 - Focus on high-revenue CPT codes.
 - Track revenue per provider and per encounter.
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10. Conclusions

Revenue depends heavily on a small number of insurance providers (mainly Medicare and Aetna) and a limited set of high-volume CPT codes. There is a consistent gap between billed and paid amounts, indicating contractual adjustments or reimbursement limitations that directly affect net revenue. Neurology carries the highest workload, suggesting strong demand but potential risk of provider strain. Some specialties, such as Cardiology, show lower utilization, indicating possible demand gaps or referral inefficiencies.